

Fig. 1: Block diagram for digital TV transmission

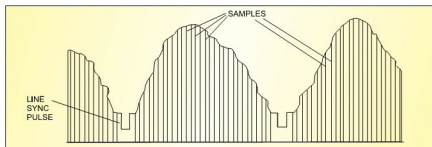


Fig. 2: Sampling of composite video signal of television

the bandwidth.

A 13.5MHz sampling rate was selected to satisfy this as well as other criteria. One is that the sampling rate must be divisible by line frequency:

$$13.5 = 864 \times 15.625 \text{ for PAL}$$

The number of active pixels per line for PAL is therefore $13.5 \times 52 = 702$

Video sampling

Colour TV broadcasting involves transmission of luminance (Y) and two colour-difference signals (R-Y and B-Y). The two colour-difference signals R-Y and B-Y are referred to as C_R and C_B respectively.

In DVB, these three components are independently sampled and converted into three digital datastreams before compression, modulation and subsequent transmission. For the luminance signal, which contains the highest video frequencies, the full sampling rate of 13.5 MHz is used. As for chrominance components C_R and C_B , which contain lower video frequencies, sampling rate of 6.75 MHz is used. It is followed by the multiplexer, where all the three streams are combined into

a single stream and a total sampling rate of $13.5 + 6.75 + 6.75 = 27$ MHz is obtained.

Sampling structure. During the digitising process, the three parameters of the component video signal are assigned a numeric sampling value. Groups of four video pixels within each of the three components are looked at and samples taken for recording. With a 4:2:2 sampled video signal, all four of the luminance pixels, two R-Y pixels and two B-Y pixels are sampled. This gives a 4:2:2 sampling rate.

With a 4:1:1 signal, all four of the luminance pixels are sampled four times but only one pixel is sampled from each of the R-Y and B-Y. This lower sampling rate of the colour components results in less colour information being recorded, affecting the accuracy and intensity of the colour in the video signal.

4:1:1 may not be used when doing chroma-keying, graphics and other compositing functions, as all of these functions require strong colours to be present in the video signal. The

advantage of 4:1:1 sampling is that you can record twice as much information as 4:2:2 on the same area of video tape, thus providing twice as much recording/playback time within a

given tape length. And, of course, the circuitry within the equipment is less expensive for a manufacturer to produce.

Bit rate

The next step after sampling is quantisation, in which sample values are rounded up to the nearest integer quantum values. The precise number of quanta or levels is determined by bit depth—the number of bits in the code. For example, if eight bits are used, there will be $2^8 = 256$ quantum levels. The bit rate can be calculated as:

Bit rate = Number of samples per second $\times$ Number of bits per sample

Number of samples per second = Number of samples per picture $\times$ Number of pictures per second

Number of samples per PAL picture = $720 \times 576 = 414,720$

Given a picture rate of 25,

Number of samples per second = $720 \times 576 \times 25 = 10,368,000$

Therefore the bit rate generated by the luminance component using an 8-bit code is:

$720 \times 576 \times 25 \times 8 = 82,944,000 = 82.944$ Mbps

The bit rate for the chrominance components depends on the sampling structure used. For a 4:2:2 sampling structure with only horizontal sub-sampling:

Number of samples = $360 \times 576 = 207,360$ per picture

Bit rate for each chrominance component is: